Week 5 Euler Totient Function

Recall the multiplicative modular arithmetic group

$$U(n) = \mathbb{Z}_n^{\times}$$
$$= \{ x \in \mathbb{Z}_n : \gcd(x, n) = 1 \}$$

x, n coprime $\iff$ x^{-1} exists modulo n

In cyclic group chapter we saw how taking an element $g \in G$ we can form the subgroup

$$\langle g \rangle = \{ g^k : k \in \mathbb{Z} \}$$

$|\langle g \rangle| = |g|$, order of element g

= least positive integer $k > 0$ such that $g^k = e$.

it's possible that no such k exists in which case we say $|g| = \infty$.

Investigate some examples of $U(n)$ and the possible orders of their elements.

First we'll look at some prime n.

$n = 13$ $\qquad |U(13)| = 12.$

$U(13) = \{1, 2, 3, \ldots, 12\}$

for x in $U(13)$ we see that

$$\boxed{|x| \in \{1, 2, 3, 4, 6, 12\}}$$

$n = 17$, $\quad |U(17)| = 16$

$x \in U(17)$, $\boxed{|x| \in \{1, 2, 4, 8, 16\}}$

$n = 31$ $\quad |U(31)| = 30$

For $x \in U(13)$ $\quad \boxed{x \in \{1, 2, 3, 5, 6, 10, 15, 30\}}$

Conjecture:

For primes p and element $x \in U(p)$ the possibilities for $|x|$ are all the factor of $p - 1 = |U(p)|$

What about some composite n.

Consider $U(100)$, $|U(100)| = 40$ (Why?)

For $x \in U(100)$

$|x|$ comes from $\{1, 2, 4, 5, 10, 20\}$.

$U(200)$, $|U(200)| = 80$

$|x|$ comes from $\{1, 2, 4, 5, 10, 20\}$

$U(500)$, $|U(500)| = 200$

$|x|$ comes from $\{1, 2, 4, 5, 10, 20, 25, 50, 100\}$.

Conjecture for composite modulus. n

If $x \in U(n)$ then $|x|$ is a factor of $|U(n)|$, but some factors might not occur.

In fact, there is Lagrange's Theorem which says:

"If $|G|$ is finite, and H is a subgroup of G, then $|H|$ is a factor of $|G|$."

ϕ "phi"

There's a special name for $|U(n)|$

The Euler totient function $\phi(n)$ is defined as

$$\phi(n) := |U(n)|$$

$$:= \text{number of } x,\ 1 \le x \le n \text{ such that } \gcd(x,n) = 1$$

- We've seen that $\phi(p) = p - 1$
- We've seen some values

$\phi(100) = 40$, $\phi(200) = 80$, $\phi(500) = 200$

But what is $\phi(n)$ in general?

<u>Theorem 5.1.</u> If $n = \prod_{i=1}^{r} p_i^{a_i}$, the prime factorization for n, i.e. $p_1, \dots, p_r$

are distinct primes and $a_i \geq 1$

<u>then</u>

$$\phi(n) = \prod_{i=1}^{r} \left(p_i^{a_i - 1} (p_i - 1) \right)$$

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eg. $\phi(100) = \phi(2^2 \cdot 5^2)$

$= 2^1 \cdot (2-1) \cdot 5^1 (5-1)$

$= 2 \cdot 5 \cdot 4 = 40$

$\phi(500) = \phi(2^2 \cdot 5^3)$

$= 2^1 (2-1) 5^2 (5-1)$

$= 2 \cdot 25 \cdot 4$

$= 200$ $\qquad f(ab) = f(a) f(b)$

<u>Proof of theorem 5.1</u> Outline.

- Already seen $\phi(p) = p - 1$
- <u>Lemma 5.2</u> $\phi(p^a) = p^{a-1}(p-1)$

- <u>Lemma 5.5</u> The function ϕ is "<u>multiplicative</u>"

ie. If $\gcd(a,b)=1$ then

$$\phi(ab) = \phi(a)\phi(b)$$

- Overall argument proceeds as.

$$\phi(n) = \phi\left(\prod_{i=1}^{r} p_i^{a_i}\right)$$

$$= \prod_{i=1}^{r} \phi(p_i^{a_i})$$, by lemma 5.5 since for $i \neq j$ $\gcd(p_i^{a_i}, p_j^{a_j}) = 1$

$$= \prod_{i=1}^{r} p_i^{a_i - 1}(p_i - 1)$$, by lemma 5.2

$\blacksquare$

Continue at 12:00.

Lemma 5.2 $\phi(p^a) = p^{a-1}(p-1)$.

Proof: Consider the range of integers.

$$1, 2, 3, \ldots\ldots \quad \ldots\ldots, p^a - 1, p^a. \quad (*)$$

By definition $\phi(p^a)$ is the number of integers in $(*)$ that are co-prime to p^a.

Let's count the ones that are not coprime to p^a.

For $1 \le x \le p^a$ consider how $\gcd(x, p^a) \neq 1$

But the only divisors of p^a are $1, p, p^2, p^3, \ldots, p^a$

So $\gcd(x, p^a) \neq 1$, $\geq p$ if and only if

$$p \mid x.$$

So how many such x are there in $(*)$? i.e. How many multiples of p in $(*)$?

They are $p, 2p, 3p, \ldots, p^a = p^{a-1} \cdot p$

so there are p^{a-1} integer x in $(*)$ such that $\gcd(x, p^a) \neq 1$.

So there are $p^a - p^{a-1}$ integers x

in (*) that $\gcd(x, p^a) = 1$

$$\Rightarrow \boxed{\phi(p^a)} = p^a - p^{a-1}$$

$$\boxed{= p^{a-1}(p-1)}$$

as required

Lemma 5.3 A divisibility exercise.

If $a \equiv b \pmod{n}$ then $\gcd(a,n) = \gcd(b,n)$

Proof comes from basic divisibility

$$a \equiv b \pmod{n} \iff n \mid a-b$$

$$\iff \exists q \in \mathbb{Z} \quad a - b = qn.$$

$$a = b + qn$$

a as a lin. comb of b and n

$$b = a - qn.$$

b as a lin. comb. of a and n.

So (a,n) & (b,n) have exactly the same common divisors

$\Rightarrow \quad \gcd(a,n) = \gcd(b,n).$ ∎

<u>Lemma 5.4</u>

For all $u, v, w \in \mathbb{Z}$ we have.

$\gcd(u, vw) = 1$

$\iff \gcd(u,v) = 1$ & $\gcd(u,w) = 1$.

<u>Proof</u>: (in notes based on lin. combs). A more straightforward argument based on the F.T.A.

$\gcd(u, vw) = 1$

$\iff (u, vw)$ share no common prime factors.

$\iff (u,v)$ share no prime factors & (u,w) " " " " .

$\iff \gcd(u,v) = 1$ & $\gcd(u,w) = 1$.

∎

Lemma 5.5 ϕ is multiplicative.

ie. $\gcd(a,b)=1 \implies \phi(ab)=\phi(a)\phi(b)$.

Proof: A counting argument, counting the m in the range $1 \le m \le ab$ that are coprime to ab

ie. $\gcd(m, ab)=1$

$\iff \gcd(m,a)=1$ & $\gcd(m,b)=1$.

, by Lemma 5.4.

Look for these m in the grid $a \times b$.

1	2	3	....	$b-1$	b
$b+1$	$b+2$	$b+3$	.....	$b+(b-1)$	$2b$
$2b+1$	$2b+2$	$2b+3$	.....		$3b$
$(a-1)b+1$	$(a-1)b+2$	$(a-1)b+3$	.....		ab
$\equiv 1 \pmod b$	$\equiv 2 \pmod b$	$\equiv 3 \pmod b$			$\equiv 0 \pmod b$

- each column is a cong. class mod b.
- in each row we see a complete set of residues/remainders modulo b.

- Claim: in each column the a entries are a complete set of residues modulo a.

<u>Proof</u>: (By contradiction)

Suppose $vb+r \equiv wb+r \pmod{a}$

$$\Rightarrow \quad vb \equiv wb \pmod{a}.$$

$$\Rightarrow \quad vbb^{-1} \equiv wbb^{-1} \pmod{a}$$

b^{-1} exists since $\gcd(a,b)=1$

$$\Rightarrow \quad v \equiv w \pmod{a}$$

But $0 \leq v, w \leq a-1$

$$\Rightarrow \quad v = w$$

Therefore each column contains a entries, which are all distinct modulo a, and hence a complete set of residues modulo a. $\blacksquare$

So now we count m from the grid that satisfy, simultaneously,

$$\gcd(m, a) = 1 \quad \& \quad \gcd(m, b) = 1$$

There are $\phi(b)$ columns, all of whose entries are coprime to b. Within any particular column, $\phi(a)$ of the entries are coprime to a.

So there are $\phi(b)\phi(a)$ of m that satisfy

$$\gcd(m, ab) = 1$$

Therefore $\boxed{\phi(ab) = \phi(a)\phi(b)}$ $\blacksquare$.

Remember $\gcd(a, b) = 1$.